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Heimdall: test-time scaling on the generative verification

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Abstract

An AI system can create and maintain knowledge only to the extent that it can verify that knowledge itself [23]. Recent work on long Chain-of-Thought reasoning has demonstrated great potential of LLMs on solving competitive problems, but their verification ability remains to be weak and not sufficiently investigated. In this paper, we propose Heimdall, the long CoT verification LLM that can accurately judge the correctness of solutions. With pure reinforcement learning, we boost the verification accuracy from 62.5% to 94.5% on competitive math problems. By scaling with repeated sampling, the accuracy further increases to 97.5%. Through human evaluation, Heimdall demonstrates impressive generalization capabilities, successfully detecting most issues in challenging math proofs, the type of which is not included during training. Furthermore, we propose Pessimistic Verification to extend the functionality of Heimdall to scaling up the problem solving. It calls Heimdall to judge the solutions from a solver model and based on the pessimistic principle, selects the most likely correct solution with the least uncertainty. Taking DeepSeek-R1-Distill-Qwen-32B as the solver model, Pessimistic Verification improves the solution accuracy on AIME2025 from 54.2% to 70.0% with 16× compute budget and to 83.3% with more compute budget. With the stronger solver Gemini 2.5 Pro, the score reaches 93.0%. Finally, we prototype an automatic knowledge discovery system, a ternary system where one poses questions, another provides solutions, and the third verifies the solutions. Using the data synthesis work NuminaMath [13] for the first two components, Heimdall effectively identifies problematic records within the dataset and reveals that nearly half of the data is flawed, which interestingly aligns with the recent ablation studies from NuminaMath.

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1 Introduction

In the realm of scientific and mathematical discovery, the process of logistic verification and validation is as crucial as the initial act of problem-solving. One of the most illustrative examples of this principle can be found in the famous thought experiment 'chasing a beam of light' by Albert Einstein, where he found the paradox within the established physics theories and further formulated the principle of the constancy of the speed of light, a cornerstone of his Special Theory of Relativity. Recently, the problem solving ability of LLMs have been significantly improved. With the long Chain of Thought(CoT) reasoning, advanced LLMs are now able to effectively solve complex competition-level problems in both math and code domains. However, the verification ability of LLMs has not been sufficiently investigated. On one hand, although the intelligence of general purposed LLM increases rapidly with the long CoT capabilities, we find that current SOTA models with direct prompting [5, 21] are not good at verifications on complex problems, e.g., o1-mini only achieves

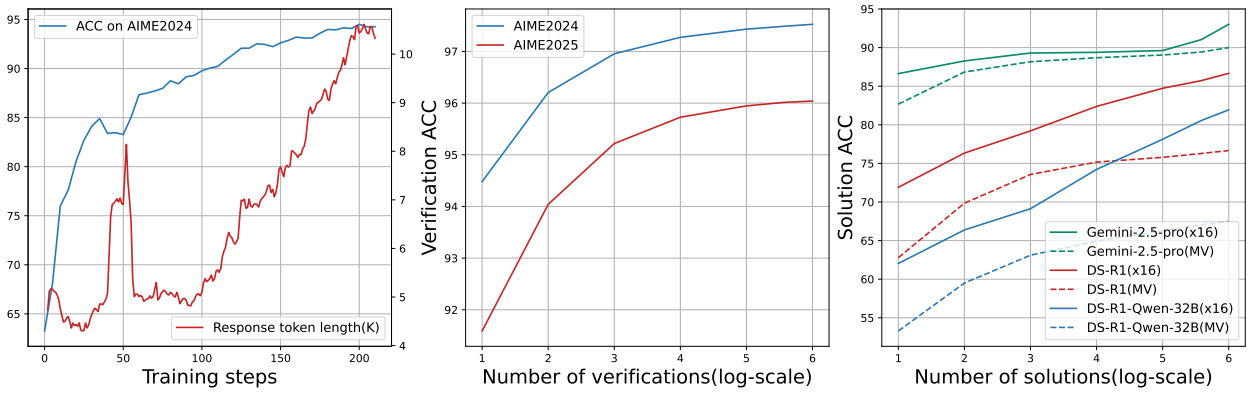

Figure 1 Scaling of Heimdall. **Left:** the verification accuracy scales with the response length during RL training. With more reasoning tokens, Heimdall gives more accurate judgment on the solutions on AIME2024. **Middle:** the verification accuracy scales with repeated sampling and Majority Voting. By sampling multiple verification trajectories and voting, the accuracy can be further improved. **Right:** with Heimdall scoring the solutions on AIME2025, the problem solving accuracy scales with the number of solutions. We verify 16 times on each solution and select the most likely correct one with Pessimistic Verification($\times$16). When inter-playing with various solver models, Heimdall gives significant improvements over pure solver-based Majority Voting(MV).

80.9% on our evaluation dataset. On the other hand, some work [16, 17, 22, 25, 29, 32, 35] trains a dedicated model for verification or critique but the high-quality verification data is hard to collect, which limits the verification capability and hence impedes the application to challenging problems.

In this paper, we claim that verifying if a solution is correct is a special type of problem solving, i.e., a true/false question and involves step-by-step judgment on the solution. Inspired by the recent progress on the long CoT reasoning, we propose to train a long CoT verifier through reinforcement learning. We name it Heimdall, symbolizing its sharp ability to detect errors and safeguards the correctness of knowledge. We leverage PPO [20] algorithm and find that the data processing is critical to the RL training. Specifically, two types of problems hinder the optimization, i.e., easy problems with only correct solutions and hard problems with only wrong solutions, both of which lack contrastive examples and tends to guide the verifier to simply identify the hardness of a problem, rather than finding the wrong position in the solution. By filtering out the two cases, the model learns the verification ability more effectively. Taking the competitive math problems as our primary experimental domain, we show that the verification ability follows the test time scaling law where the accuracy improves significantly from 62.5% to 94.5% as the response length grows, as is shown the left of Figure 1. The performance can be further improved by sampling multiple verifications and voting on the judgment results. In the middle of Figure 1, the accuracy grows from 94.5% to 97.5% on AIME2024 as the number of verifications grows from 2 to 64. Furthermore, the evaluation from human experts shows that Heimdall generalizes well on math proof problems although it is trained with only the calculation problems with explicit answers.

In addition, we extend the usage of Heimdall to scale up the problem solving. Suppose the solver model gives multiple solutions for a problem and Heimdall judges the correctness of each solution for multiple times. We can select the best solution based on the verification results. We frame the selection process as a multi-arm bandit problem where solutions with the same conclusion are treated as multiple visits to the same 'arm'. Based on the pessimism principle, we propose the solution selection algorithm called Pessimistic Verification that minimizes the uncertainty of selecting wrong solutions. The algorithm unifies Majority Voting and reward model based Best-of-N by balancing the contributions of the solver and the verifier, and empirically demonstrates better scaling over both algorithms. Taking DeepSeek-R1-Distill-Qwen-32B [6] as the solver model, which scores 54 on AIME2025, Pessimistic Verification raises the scores to 70 with $\times$16 compute, matching the performance of o1, and to 83.3 with more compute. We further test with stronger solver models, including DeepSeek-R1 [6] and Gemini 2.5 Pro[3]. As is shown in the right of Figure 1, Pessimistic Verification with Heimdall consistently improves the problem solving of various models and with Gemini 2.5 Pro, the

accuracy on AIME2025 reaches 93%, matching the currently reported SOTA with multiple attempts by Grok3 [4].

Finally, we create a prototype to demonstrate the utility of Heimdall on the automatic knowledge discovery. We use the work of math data synthesis called NuminaMath [12] as the procedure of automatically proposing new problems and the corresponding solutions, and call Heimdall to detect errors in the synthetic dataset. Human annotation demonstrates that Heimdall accurately identifies the errors in the dataset. The verification result also reveals that the quality of current synthetic dataset is poor, which is consistent with the authors' finding that removing the dataset from training improve the performance of the solver model [13].

In summary, our contributions are as follows:

- We propose Heimdall, the long CoT verifier by reinforcement learning and demonstrate the superior accuracy than top-tier LLMs. Heimdall also shows good generalization ability on out-of-domain problems, such as math proof problems.
- We propose a unified algorithm called Pessimistic Verification for inference time scaling on problem solving. Empirically, it scales better than the vanilla Majority Voting or the reward-model based Best-of-N and achieve SOTA accuracy on AIME2025.
- We create a prototype to show the utility of Heimdall in the autonomous knowledge discovery, where Heimdall is used to identify the correctness of the problem-solution pairs synthesized by another LLM. Human evaluations shows that Heimdall can effectively detect the flaws in the synthetic data.

2 Related Work

Reasoning model. Reasoning models outperform previous general-purpose models on challenging reasoning tasks. During the chain of thought(CoT) reasoning, they keep reflecting their claims and searching viable solutions, utilizing more compute budget and providing better and more robust results. OpenAI first released its reasoning models[2, 7, 30] that performs significantly better on competitive tests like AIME and CodeForces than its previous models. Work by DeepSeek[6] and Kimi[24] independently propose different ways of reinforcement learning to trigger the reflection and searching capability in their base models. Recently, Grok3 [4] and Gemini 2.5 Pro [3] also demonstrate their impressive reasoning capabilities through long CoT.

Generative evaluation. Recently some work are interested in improving the verification ability of the LLMs. Some [8, 29, 32, 34] explores finetuning an LLM with synthetic verification data to improve its verification ability. However, it is hard to synthesize high-quality data if the LLM inherently lacks the verification skills. One of the related topic is LLM-as-a-Judge [5, 11] where a LLM is prompted to evaluate responses from other LLMs. The work[11] leverages strong LLMs as judges to evaluate other models in various domains and reveals that strong LLM judges have good generalization ability in different domains. Some works design the judge system and analyze of the judgment behavior[19, 21, 27]. However, prompting is only effective on easy tasks, and when it comes to competitive tests, the general purposed reasoning models performs not quite well, as is reveals from our test in Section 4. Another similar topic is critique [9, 10, 14, 15, 18], which often focuses on code and math problems and is used for giving suggestions for further revision. Critique fine-tuning [25] shows that fine-tuning on a high-quality critique data is beneficial to the reasoning ability of a base model. Several work, e.g., CTRL [26] leverages RL to train LLMs. However, they do not leverage the long CoT ability, which limits the verification performance on complex reasoning problems.

3 Approach

We define the verification task where we ask a model to judge if a solution to a problem is correct in its CoT and finally put its judgment result at the end of the response. Table 1 is the template of the verification prompt and the expected format of a response.